

Lesson 03: Student Study Guide

Learn With Real Projects - Vision Labs

1. Executive Summary

This study guide covers Python conditional logic (if, elif, else), logical operators (and, or, not), comparison operators, and nested decision-making structures. Students will understand how to evaluate Boolean expressions in isolation and how to apply these rules to build an automated CRM Lead Qualification and SLA Routing Engine for Vision Labs.

2. Core Technical Concepts

- Comparison Operators (>, <, >=, <=, ==, !=): Return pure Boolean values (True or False) based on value evaluation.
- Multi-Way Branching (elif): Evaluates mutually exclusive conditions sequentially until a True branch is found.
- Logical Operators (and, or, not): Combine multiple Boolean tests or invert boolean evaluation.
- Guard Clauses / Early Exit Pattern: Aborts function execution early for unqualified or invalid data to maintain code efficiency.
- Operator Precedence: Using parentheses to control evaluation order in complex logical expressions.

3. Practice Code Breakdown (practice/exercise_01.py)

The practice exercise breaks down syntax into four clear blocks:

1. Basic Comparison: Checks if `client_budget >= 5000` to trigger a simple two-way decision.
2. Multi-Way ELIF: Classifies `lead_score` sequentially into Enterprise, Mid-Market, SMB, or Unqualified.
3. Logical Operators: Demonstrates how 'and' requires both conditions, 'or' requires one condition, and 'not' flips Booleans.
4. Nested Logic: Places an inner deadline check inside an outer VIP status check.

4. Production Code Architecture (main.py)

The `main.py` script implements the `evaluate_lead()` function for Vision Labs:

- Inputs: Accepts client details with static Python type hinting.
- Compound Lead Tiering: Evaluates budget and existing client status using combined 'or' and 'and' logic.
- Guard Clause: Returns early with a 'REJECTED' payload for unqualified leads.
- Dynamic SLA Routing: Uses nested conditionals to map qualified leads to queues like `CRITICAL_RUSH_ENGINEERING` or `SENIOR_FULLSTACK_QUEUE`.
- Discount Logic: Protects business margins by ensuring rush deadlines (`<= 3` days) are excluded from promotional discounts.

5. Key Takeaways & Student Exercise Challenge

STUDENT CHALLENGE:

Modify `evaluate_lead()` in `main.py` to add a new parameter: `'is_non_profit'` (bool). If a lead is a non-profit organization AND their budget is at least \$2,000, automatically upgrade their tier to 'Tier 2 - Mid-Market' and grant them discount eligibility regardless of their deadline. Test your updated logic using a new lead entry in the execution driver.